

CS312 MID TERM Exam Objective Part Module 01 to 11

These MCQs and explanations cover the key concepts of the module and should be useful for your exam preparation. . .

MCQs for Module-01: Introduction to Data & Information

1. What is the primary difference between data and information?

- A. Data is structured, while information is unstructured.
- B. Data is raw and unprocessed, while information is processed and organized.
- C. Data is always numerical, while information is always textual.
- D. Data is useful for decision making, while information is not.

Answer: B

2. Which of the following statements is true about data?

- A. Data is always ready for decision making.
- B. Data has an internal structure.
- C. Data is the same as information.
- D. Data does not require further analysis.

Answer: B

3. When does data become information?

- A. When it is collected.
- B. When it is stored in a computer.
- C. When it is processed and interpreted in a given context.
- D. When it is used to make decisions without any processing.

Answer: C

4. Which of the following is an example of data?

- A. A list of students with CGPA more than three.
- B. A file containing names and phone numbers in a telephone directory.

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- C. A report summarizing student performance.
- D. A chart showing monthly sales trends.

Answer: B

5. What is the role of data in decision making?

- A. Data is directly used to make decisions without any further processing.
- B. Data provides a foundation that is analyzed to generate information suitable for decision making.
- C. Data and decision making are unrelated.
- D. Data complicates the decision-making process.

Answer: B

6. How is a telephone directory classified in terms of data and information?

- A. It is considered information because it is organized.
- B. It is considered data because it is a large collection of facts with structure.
- C. It is neither data nor information.
- D. It is considered information because it includes names and addresses.

Answer: B

7. In the context of student records on a computer, what transforms the data into information?

- A. Adding more students to the file.
- B. Sorting the students by name.
- C. Processing the data to get a list of students with a CGPA more than three.
- D. Deleting records of graduated students.

Answer: C

8. Which of the following best describes 'information'?

- A. Raw facts and figures.
- B. Processed data that is meaningful and useful for decision making.
- C. Unstructured data sets.
- D. A collection of random numbers.

Answer: B

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9. What is meant by saying "information is a subset of data"?

- A. Information includes all the data available.
- B. Information is a smaller, specific portion of data that is processed.
- C. Information and data are identical in every way.
- D. Information does not depend on data.

Answer: B

10. Which example demonstrates the transformation of data into information?

- A. Recording student names in a file.
- B. Creating a database of telephone numbers.
- C. Generating a list of top-performing students based on CGPA.
- D. Collecting survey responses.

Answer: C

11. What qualifies a file storing flight routes from one city to another as data?

- A. It contains information about specific flights.
- B. It is structured with details like Flight ID, Origin, Destination, Date of Departure, and Time of Departure.
- C. It is processed to retrieve flight information for specific destinations.
- D. It is used directly for decision making.

Answer: B

12. How does data become information in the context of flight routes?

- A. When the file is organized into a database.
- B. When the file is filtered to get details of flights for a specific route and date.
- C. When the flight details are printed out.
- D. When the data is stored in a file.

Answer: B

13. What is a key benefit of using databases?

- A. Increased data redundancy.
- B. Manual handling of large data sets.

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- C. Reduced risk of data loss and ease of data access.
- D. Increased complexity in data retrieval.

Answer: C

14. Why is a database not called an information base?

- A. Because it only stores unorganized data.
- B. Because databases only deal with optimizing data storage, not information retrieval.
- C. Because databases do not store any information.
- D. Because information bases are a different technology.

Answer: B

15. What distinguishes a Database from a Database Management System (DBMS)?

- A. Database is a tool for data manipulation, DBMS is just data storage.
- B. Database is the data structure, DBMS is the system to manage and retrieve the data.
- C. Database is software, DBMS is hardware.
- D. Database is used for decision making, DBMS is not.

Answer: B

16. Which of the following is an example of a DBMS?

- A. A file with employee records.
- B. Oracle software used to manage employee records.
- C. A hard disk storing individual files.
- D. A report generated from employee data.

Answer: B

17. What are the components of a Database Management System (DBMS)?

- A. Data, Users, Hardware, Software, Procedures
- B. Data, Files, Software, Printers, Network
- C. Data, Users, Network, Printers, Software
- D. Users, Hardware, Network, Printers, Data

Answer: A

18. What role do procedures play in a DBMS?

- A. They store data in the database.
- B. They refer to instructions and rules for designing and using the database.
- C. They provide the hardware for the DBMS.
- D. They are used to write software for the DBMS.

Answer: B

19. In the context of databases, what is meant by 'reducing redundancy'?

- A. Eliminating unnecessary duplicate data.
- B. Increasing the number of backup copies.
- C. Adding more data to the database.
- D. Ensuring data is frequently updated.

Answer: A

20. What is the function of the hardware component in a DBMS?

- A. To manage and control the overall database.
- B. To provide physical electronic devices for data storage and retrieval.
- C. To write and execute database procedures.
- D. To design the database structure.

Answer: B

MCQs for Module 02: Data Storage Mechanism

1. What is a data model in the context of databases?

- A. A system for backing up data
- B. A basic architectural unit that determines the structure, connection, processing, and storage of data
- C. A software tool for data visualization
- D. A method for encrypting database information

Answer: B

2. Which of the following statements is true about a flat file database?

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- A. It supports complex relationships between data elements.
- B. It contains one record per line and is stored as an ordinary file on its host computer.
- C. It uses a relational model to connect data.
- D. It minimizes data redundancy.

Answer: B

3. How is data stored in a flat file database?

- A. In a hierarchical structure with parent and child nodes
- B. In multiple interconnected tables
- C. In columns and rows, with each row separated by a comma or tab
- D. In an encrypted format to ensure security

Answer: C

4. What is a major disadvantage of using a flat file database?

- A. It supports many-to-many relationships.
- B. It has high data integrity.
- C. It results in data duplication and reduced integrity.
- D. It uses complex querying mechanisms.

Answer: C

5. In a hierarchical database model, how is data retrieved?

- A. By using complex SQL queries
- B. By visiting the root node and navigating through its children
- C. By referencing addresses of all the records
- D. By direct access using a unique identifier

Answer: B

6. Which relationship is true for a hierarchical database model?

- A. Many-to-many between root and child nodes
- B. One-to-many between parent and child nodes
- C. One-to-one between all nodes
- D. Many-to-one between child nodes and root

Answer: B

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7. How does a network database model differ from a hierarchical model?

- A. It supports one-to-many relationships exclusively.
- B. It allows each record to have multiple parents and child records.
- C. It stores data in a flat structure.
- D. It cannot represent complex relationships.

Answer: B

8. What is a key characteristic of a network database model?

- A. It uses a flat file structure.
- B. It stores data in a hierarchical manner.
- C. It supports many-to-many relationships and forms a generalized graph structure.
- D. It lacks flexibility in representing data relationships.

Answer: C

9. In a network database, how are records linked?

- A. Through a strict parent-child hierarchy
- B. Using addresses of all the referencing records, allowing any record to link to any other record
- C. By storing records in a single table without any relationships
- D. By using primary keys exclusively

Answer: B

10. What is a major benefit of the network database model?

- A. It simplifies the database design by avoiding relationships.
- B. It is easy to navigate due to linkages and many-to-many relationships.
- C. It eliminates the need for data backup.
- D. It reduces the need for data validation.

Answer: B

MCQs for Module 03: Three-Tier Architecture

1. What is a major advantage of using a 3-Tier Architecture?

- A. It simplifies the entire application into a single layer.
- B. It allows different teams to work on different layers simultaneously.
- C. It eliminates the need for a database.
- D. It reduces the need for an application interface.

Answer: B

2. What does the Presentation Layer in 3-Tier Architecture manage?

- A. Data storage
- B. Business logic
- C. Input/output data and their display
- D. Database connections

Answer: C

3. Which layer in 3-Tier Architecture contains the business and data rules?

- A. Presentation Layer
- B. Application Tier (Business Logic)
- C. Data Tier
- D. Client Interface

Answer: B

4. In 3-Tier Architecture, what is the role of the Data Tier?

- A. Managing user interfaces
- B. Storing information and optimizing data access
- C. Processing business logic
- D. Handling client requests directly

Answer: B

5. How do the layers in 3-Tier Architecture interact with each other?

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- A. Each layer directly interacts with every other layer.
- B. Layers are completely independent and do not interact.
- C. Each layer interacts only with its adjacent layers.
- D. All layers interact through a central control system.

Answer: C

6. What is a key benefit of Layer Independence in 3-Tier Architecture?

- A. All layers share the same logic and data.
- B. Changes in one layer do not affect the other layers.
- C. The presentation layer can directly manipulate the data layer.
- D. It simplifies the application to a single monolithic unit.

Answer: B

7. Which role is responsible for the physical realization and maintenance of the database in a DBMS environment?

- A. Application Programmer
- B. End User
- C. Database Administrator (DBA)
- D. System Analyst

Answer: C

8. What does Client-Server Architecture primarily involve?

- A. Clients providing services to the server
- B. Servers requesting services from clients
- C. Clients requesting services from a centralized server
- D. Distributed servers communicating directly with each other

Answer: C

9. In the context of Client-Server Architecture, what is the main role of the server?

- A. To provide an interface for users
- B. To wait for and respond to requests from clients
- C. To act as a client to other servers
- D. To store client-side data

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Answer: B

10. How does Distributed Processing enhance computing power?

- A. By centralizing all processing tasks in one system
- B. By distributing processing tasks across multiple systems
- C. By reducing the number of messages passed between systems
- D. By eliminating the need for network communication

Answer: B

MCQs for Module 04: Fact Finding Technique

1. Which fact-finding technique involves examining existing documents to understand the current system and identify data requirements?

- A. Conducting Interviews
- B. Observing the Enterprise in Operations
- C. Examine Documents
- D. Research

Answer: C

2. What is a key advantage of using the document examination technique in database development?

- A. It directly gathers user opinions
- B. It helps identify data relationships and information flow within the system
- C. It allows for real-time observation of activities
- D. It involves the end user in generating enthusiasm for the project

Answer: B

3. What is the primary purpose of conducting interviews in the context of fact-finding?

- A. To observe the actual operations of an enterprise
- B. To gather detailed and diverse information directly from stakeholders
- C. To examine existing documents for data relationships
- D. To perform research using secondary sources

Answer: B

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4. Which type of interview allows the interviewee to guide the conversation and provide direction?

- A. Structured Interview
- B. Unstructured Interview
- C. Open-ended Interview
- D. Close-ended Interview

Answer: B

5. What type of questions does a structured interview typically use?

- A. Open-ended questions only
- B. Close-ended questions only
- C. Both open-ended and close-ended questions
- D. Free-format questions

Answer: C

6. Which fact-finding technique is particularly useful when the system is too complex to be explained by end users?

- A. Conducting Interviews
- B. Observing the Enterprise in Operations
- C. Examine Documents
- D. Questionnaire

Answer: B

7. During observation, what is crucial for the observer to avoid in order to get accurate information?

- A. Taking notes during the observation
- B. Asking too many questions while the activity is being performed
- C. Observing the activity during peak periods only
- D. Interacting with multiple users simultaneously

Answer: B

8. What is one of the main advantages of using a questionnaire for fact-finding?

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- A. It allows for real-time data collection
- B. It can gather information from a large number of people
- C. It involves detailed face-to-face interaction
- D. It ensures observation of actual operations

Answer: B

9. Which type of questionnaire question offers respondents greater freedom in answering?

- A. Fixed-Format Questions
- B. Close-ended Questions
- C. Free-Format Questions
- D. Multiple-Choice Questions

Answer: C

10. What is a disadvantage of using fixed-format questions in a questionnaire?

- A. They provide subjective responses that are difficult to tabulate
- B. They limit the responses to specific options, potentially missing additional information
- C. They allow respondents to answer in any way they choose
- D. They require detailed and lengthy responses

Answer: B

MCQs for Module 05: Implementing Fact Finding Techniques

1. What is the immediate step after fact-finding in the database design cycle?

- A. Conducting Interviews
- B. Requirement Writing
- C. System Definition
- D. Database Implementation

Answer: B

2. Which document describes the data to be held in a database and how the data is to be used, including the processes and relationships?

- A. Interview Transcript
- B. Questionnaire

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C. Specification Document

D. Observation Report

Answer: C

3. In the given interview example for a rental system, what specific data does the interviewee mention storing about the customer?

A. Customer ID, name, phone, and email

B. Customer ID, name, phone, and address

C. Customer ID, name, address, and email

D. Customer name, address, and email

Answer: B

4. From the example of translating a questionnaire into requirements, what is the targeted price range for the dishes offered?

A. 500-1000

B. 700-1200

C. 1000-1500

D. 1200-1500

Answer: B

5. In the restaurant management system requirements, how is the cost of each dish calculated?

A. Cost of ingredient only

B. Cost of ingredient and electricity

C. Cost of ingredient plus % of profit

D. Fixed price based on market rates

Answer: A

6. What percentage of profit is added to the cost of ingredients in the restaurant management system example?

A. 10%

B. 15%

C. 20%

D. None of the above

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Answer: C

7. What is the main purpose of the system definition phase in database design?

- A. To gather all necessary data
- B. To define the detailed architecture and boundaries of the system
- C. To conduct interviews and observations
- D. To implement the database

Answer: B

8. What should be included in the system definition for a database system?

- A. Initial database size
- B. Database rate of growth
- C. Performance and security measures
- D. All of the above

Answer: D

9. Which activity forms the foundation of the system definition by examining the needs and requirements of stakeholders?

- A. Requirement Writing
- B. Concept Definition
- C. Database Implementation
- D. Fact-Finding

Answer: B

10. During the process of extracting system definition, what method is used to divide the system into subsystems?

- A. Bottom-up aggregation
- B. Lateral decomposition
- C. Top-down decomposition
- D. Horizontal integration

Answer: C

MCQs for Module 06: Process to Database Design

1. What is the primary purpose of the requirement gathering process?

- A. To implement the database
- B. To identify the actual need and purpose of a system
- C. To design the user interface
- D. To test the system

Answer: B

2. What is a major challenge in requirement documentation or gathering?

- A. Implementing new features
- B. Managing system updates
- C. Missing or wrong requirements
- D. System deployment

Answer: C

3. Which attribute of software requirements ensures they are measurable and testable?

- A. Completeness
- B. Traceability
- C. Consistency
- D. Verifiability

Answer: D

4. What is the meaning of completeness in software requirements?

- A. Requirements are non-conflicting
- B. Requirements can be verified
- C. Requirements cover all real-world situations and system responses
- D. Requirements have unique identifiers

Answer: C

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5. What does consistency in software requirements imply?

- A. Requirements are easy to verify
- B. Requirements are free from conflicts
- C. Requirements are fully detailed
- D. Requirements are changeable

Answer: B

6. What does traceability in software requirements refer to?

- A. Requirements can be verified
- B. Requirements are unique and can be tracked
- C. Requirements are consistent
- D. Requirements are complete

Answer: B

7. What is the primary focus of requirements engineering?

- A. Coding and implementation
- B. Testing and deployment
- C. Identifying and communicating the purpose of a software-intensive system
- D. Database design

Answer: C

8. In the context of requirements management, what is a baseline requirement?

- A. A requirement that has been verified
- B. A requirement that is consistent
- C. A requirement that serves as a foundation for further development
- D. A requirement that has been traced

Answer: C

9. What is the main goal of requirement verification?

- A. To collect requirements
- B. To ensure requirements are correct and complete

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- C. To develop the system
- D. To manage project scope

Answer: B

10. What does the term "user friendliness" in quality metrics refer to?

- A. System reliability
- B. System robustness
- C. Ease of use and learnability
- D. System performance

Answer: C

11. What is the purpose of requirement analysis?

- A. To implement the system
- B. To break down requirements for a clear development process
- C. To design the user interface
- D. To verify the system requirements

Answer: B

12. During requirement prioritization, how are requirements typically categorized?

- A. Mandatory, optional, and unnecessary
- B. High, medium, and low priority
- C. Functional and non-functional
- D. Simple and complex

Answer: B

13. What is the result of the requirement specification phase?

- A. A list of system bugs
- B. A software requirement specification document
- C. A user interface design
- D. A project timeline

Answer: B

14. In the example of a quality metric for robustness, what is being evaluated?

- A. System performance under normal conditions
- B. System behavior with abnormal inputs
- C. System reliability over time
- D. System user interface design

Answer: B

15. What is a defect in the context of requirement verification?

- A. Noncompliance with agreed behavior
- B. A system feature not working as expected
- C. Wrong definition of client's requirements
- D. A minor system bug

Answer: C

Module 07: Multiple-Choice Questions MCQS and Answers

1. According to Codd's Rule No. 01, how is all information represented in a relational database?

- A) In arrays
- B) In multiple formats
- C) By values in tables
- D) By values in files

Answer: C) By values in tables

2. What does Codd's Rule No. 02 emphasize regarding data access in a relational database?

- A) Logical accessibility of data
- B) Physical accessibility of data
- C) Access through graphical interfaces only
- D) Random accessibility of data

Answer: A) Logical accessibility of data

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3. How does Codd's Rule No. 03 define the handling of NULL values in a relational database?

- A) As zero values
- B) As empty character strings
- C) As missing information
- D) As non-existent data

Answer: C) As missing information

4. According to Codd's Rule No. 04, where must the database description be available?

- A) In offline catalogs
- B) In separate files
- C) In online catalogs
- D) In external databases

Answer: C) In online catalogs

5. What characteristic must a language have according to Codd's Rule No. 05 to access a database?

- A) Non-linear syntax
- B) Graphical syntax
- C) Linear syntax
- D) Conditional syntax

Answer: C) Linear syntax

6. Which rule deals with the systematic handling of views in a relational database?

- A) Codd's Rule No. 01
- B) Codd's Rule No. 06
- C) Codd's Rule No. 09
- D) Codd's Rule No. 12

Answer: B) Codd's Rule No. 06

7. What does Codd's Rule No. 08 emphasize regarding the physical storage of data?

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- A) It must be independent of logical access
- B) It must be directly accessible to users
- C) It must require changes to the application
- D) It must rely on the logical structure

Answer: A) It must be independent of logical access

8. According to Codd's Rule No. 10, what must be independent of the application using the database?

- A) Data access
- B) Integrity constraints
- C) Data manipulation
- D) Schema design

Answer: B) Integrity constraints

9. What does Codd's Rule No. 11 state regarding the manipulation of data in a relational database?

- A) Data should not be manipulated
- B) Data manipulation must be centralized
- C) Data manipulation must be distributed
- D) Data manipulation must be supervised

Answer: C) Data manipulation must be distributed

10. Which language is commonly used for managing data in a relational database management system (RDBMS)?

- A) Python
- B) Java
- C) C++
- D) SQL

Answer: D) SQL

Module 08: Multiple-Choice Questions MCQS and Answers

1. Who developed the entity-relationship modeling and in which year was it published?

- A) Edgar F. Codd, 1970
- B) Peter Chen, 1976
- C) Charles Bachman, 1960

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- D) E. F. Codd, 1979

Answer: B) Peter Chen, 1976

2. What does an Entity Relationship Diagram (ERD) primarily represent?

- A) The business process
- B) The relationships between data
- C) The user interface
- D) The software architecture

Answer: B) The relationships between data

3. What are the three main components of an ERD?

- A) Entities, Attributes, Constraints
- B) Entities, Relationships, Diagrams
- C) Entities, Attributes, Relationships
- D) Tables, Columns, Relationships

Answer: C) Entities, Attributes, Relationships

4. In an ERD, what does an 'entity' represent?

- A) A database table
- B) A real-world object that can be identified
- C) A user of the database
- D) A function of the system

Answer: B) A real-world object that can be identified

5. Which of the following is an example of an entity in a school database?

- A) Name
- B) Age
- C) Student
- D) Class

Answer: C) Student

6. What term is used for the smallest storage unit of any database?

- A) Entity
- B) Attribute
- C) Relationship
- D) Tuple

Answer: B) Attribute

7. How is the interaction among entities captured in an ERD?

- A) Through keys
- B) Through relationships
- C) Through constraints
- D) Through triggers

Answer: B) Through relationships

8. What is the third step in the process of creating an ERD?

- A) Define entities
- B) Identify attributes
- C) Chalk out relationships
- D) Identify cardinalities

Answer: D) Identify cardinalities

9. What is the term used for the number of entities in an entity-set associated with another entity-set?

- A) Entity set
- B) Relationship set
- C) Cardinality
- D) Attribute set

Answer: C) Cardinality

10. Which of the following describes a "Many-to-Many" relationship?

- A) One entity in set A is related to one entity in set B
- B) One entity in set A is related to multiple entities in set B
- C) Multiple entities in set A are related to multiple entities in set B

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- D) One entity in set A is not related to any entity in set B

Answer: C) Multiple entities in set A are related to multiple entities in set B

Module 09: Multiple-Choice Questions MCQS and Answers:

- What does a primary key do in a relational table?
 - A) Uniquely identifies each record
 - B) Describes the attributes of an entity
 - C) Establishes foreign key constraints
 - D) Indicates optional relationships
 - Answer: A) Uniquely identifies each record
- A super key is:
 - A) A single attribute that uniquely identifies a record
 - B) A set of attributes that can uniquely identify a record
 - C) Always the same as the primary key
 - D) A foreign key in another table
 - Answer: B) A set of attributes that can uniquely identify a record
- Which of the following can have NULL values?
 - A) Primary Key
 - B) Unique Key
 - C) Super Key
 - D) Candidate Key
 - Answer: B) Unique Key
- A composite primary key:
 - A) Consists of a single attribute
 - B) Combines more than one attribute to uniquely identify a record
 - C) Is always a surrogate key
 - D) Cannot be a foreign key
 - Answer: B) Combines more than one attribute to uniquely identify a record
- In a one-to-many relationship:
 - A) Each record in the first table can have multiple matching records in the second table
 - B) Each record in the first table can have only one matching record in the second table

- C) Each record in the second table can have multiple matching records in the first table
 - D) Both A and C
 - **Answer:** A) Each record in the first table can have multiple matching records in the second table
-
- What is the purpose of a foreign key?
 - A) To uniquely identify a record in a table
 - B) To establish a relationship between two tables
 - C) To serve as a primary key in the parent table
 - D) To indicate optional relationships
 - **Answer:** B) To establish a relationship between two tables
-
- What does the term "cascading delete" refer to?
 - A) Deleting records from the child table automatically when the parent record is deleted
 - B) Deleting records from the parent table automatically when the child record is deleted
 - C) Updating records in the child table automatically when the parent record is updated
 - D) Restricting delete operations to prevent referential integrity violations
 - **Answer:** A) Deleting records from the child table automatically when the parent record is deleted
-
- Which of the following statements about surrogate keys is true?
 - A) They are always composite keys
 - B) They are natural keys
 - C) They are artificially produced and usually system-generated
 - D) They must be a combination of foreign keys
 - **Answer:** C) They are artificially produced and usually system-generated

Module 10: Multiple-Choice Questions MCQS and Answers:

What is the main purpose of the Extended Entity Relationship Diagram (EERD)?

- A) To create complex data types
- B) To avoid redundancy by sharing common attributes
- C) To simplify relational databases

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D) To enhance the data security

Answer: B) To avoid redundancy by sharing common attributes

In an EERD, what is a super type?

A) An entity type with unique attributes

B) An entity type that shares common attributes with subtypes

C) A subtype with additional relationships

D) A database schema element

Answer: B) An entity type that shares common attributes with subtypes

Which of the following best describes a subtype in an EERD?

A) An entity that contains common attributes of other entities

B) A derived class that inherits attributes and relationships from a super type

C) An isolated entity with no relationships

D) A relational table with foreign keys

Answer: B) A derived class that inherits attributes and relationships from a super type

Which rule states that every instance of the super type must also be an instance of one of the subtypes?

A) Exclusive rule

B) Inclusion rule

C) Exhaustive rule

D) Dependency rule

Answer: C) Exhaustive rule

What does the mutually exclusive property in an EERD imply?

A) Subtypes can share attributes and relationships

B) Subtypes maintain isolation of attributes and relationships

C) All subtypes are visible to each other

D) Super type can access attributes of subtypes

Answer: B) Subtypes maintain isolation of attributes and relationships

If a super type `Staff` has two subtypes `Full-time` and `Part-time`, and `Full-time` has a unique attribute `Salary`, what would violate the Exhaustive rule?

- A) Adding another attribute to `Full-time`
- B) Removing the attribute `Salary` from `Full-time`
- C) Adding a common attribute to the super type
- D) Creating another subtype without unique attributes

Answer: B) Removing the attribute `Salary` from `Full-time`

In the context of an EERD, what does a dotted line typically represent?

- A) Optionality
- B) Mandatory participation
- C) Super type relationship
- D) Cardinality

Answer: A) Optionality

Which of the following is true about the relationship between super type and subtypes?

- A) Super type can inherit attributes from subtypes
- B) Subtypes inherit attributes from the super type
- C) Both super type and subtypes share attributes mutually
- D) Super type must have unique attributes that subtypes do not have

Answer: B) Subtypes inherit attributes from the super type

What is the result of making a sub type mutually exclusive in an EERD?

- A) It can share attributes with other sub types
- B) It cannot have any unique attributes
- C) It maintains isolated attributes and relationships
- D) It can be accessed by other sub types

Answer: C) It maintains isolated attributes and relationships

In the provided HR system example, which type of employee can avail the facility of a loan?

- A) Part-time employees
- B) Full-time employees
- C) Contractual employees
- D) All employees

Answer: B) Full-time employees

Module 11: Multiple-Choice Questions MCQS and Answers

• **In the Building-Apartment renting scenario, what is the relationship between a customer and an apartment?**

- A) One-to-Many
- B) Many-to-Many
- C) One-to-One
- D) Many-to-One

Answer: D) Many-to-One

• **In the Building-Apartment renting scenario, which attribute uniquely identifies an apartment?**

- A) Building Name
- B) Apartment ID
- C) Customer ID
- D) Floor Number

Answer: B) Apartment ID

• **Which of the following is an example of an attribute for the Customer entity in the Building-Apartment renting scenario?**

- A) Floor Number
- B) Apartment ID
- C) Contact
- D) Status

Answer: C) Contact

• In the musical bands scenario, what is the maximum number of songs an album can contain?

- A) 1
- B) 6
- C) 10
- D) 12

Answer: D) 12

• Which entity is associated with the attribute 'charges' in the musical bands scenario?

- A) Album
- B) Writer
- C) Company
- D) Band

Answer: B) Writer

• In the airport management system scenario, which entity includes the attribute 'annual traffic'?

- A) Flight
- B) Airport
- C) Passenger
- D) Manager

Answer: B) Airport

• In the airport management system scenario, which entity represents the relationship between planes and flights?

- A) Passenger
- B) Plane_flight
- C) Ticket
- D) Manager

Answer: B) Plane_flight

• In the Building-Apartment renting scenario, how is the relationship between floors and buildings represented in the physical data model?

- A) Floor (floor #, no_apt, bname)
- B) Apartment (aid, status, floor#, bname, CA, cid)
- C) Building (bname, address)
- D) Customer (cid, name, address, contact#)

Answer: A) Floor (floor #, no_apt, bname)

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• In the musical bands scenario, which table includes the foreign key attributes 'bname' and 'aname'?

- A) Band
- B) Company
- C) Songs
- D) Album

Answer: C) Songs

In the airport management system scenario, what is the role of 'eid' in the Manager table?

- A) Primary key
- B) Foreign key
- C) Composite key
- D) Unique key

Answer: A) Primary key

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